Evolution of steroid 5-alpha reductases included multiple horizontal gene transfers

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Steroid 5-alpha reductases are a family of enzymes with conserved structure and activity in plants and vertebrates. In humans, this enzyme is an **important pharmacological target** for treating hormone-sensitive neoplasms. Both in plants and vertebrates, they act in **steroid hormone synthesis**. The **origin of this similarity** requires evolutionary characterization: it is worth understanding whether this is a case of **convergent acquisition** of the enzyme by horizontal gene transfers or this is a case of inheritance from a **common ancestor**.

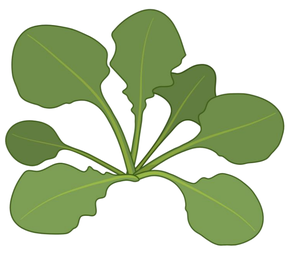
I have constructed **phylogenetic trees** of plant, vertebrate, and prokaryotic steroid 5-alpha reductases by neighbor joining method (MAFFT online) and minimum evolution method (MEGA). The trees were visualized in iTOL. Below, the **neighbor joining tree** is depicted and discussed.

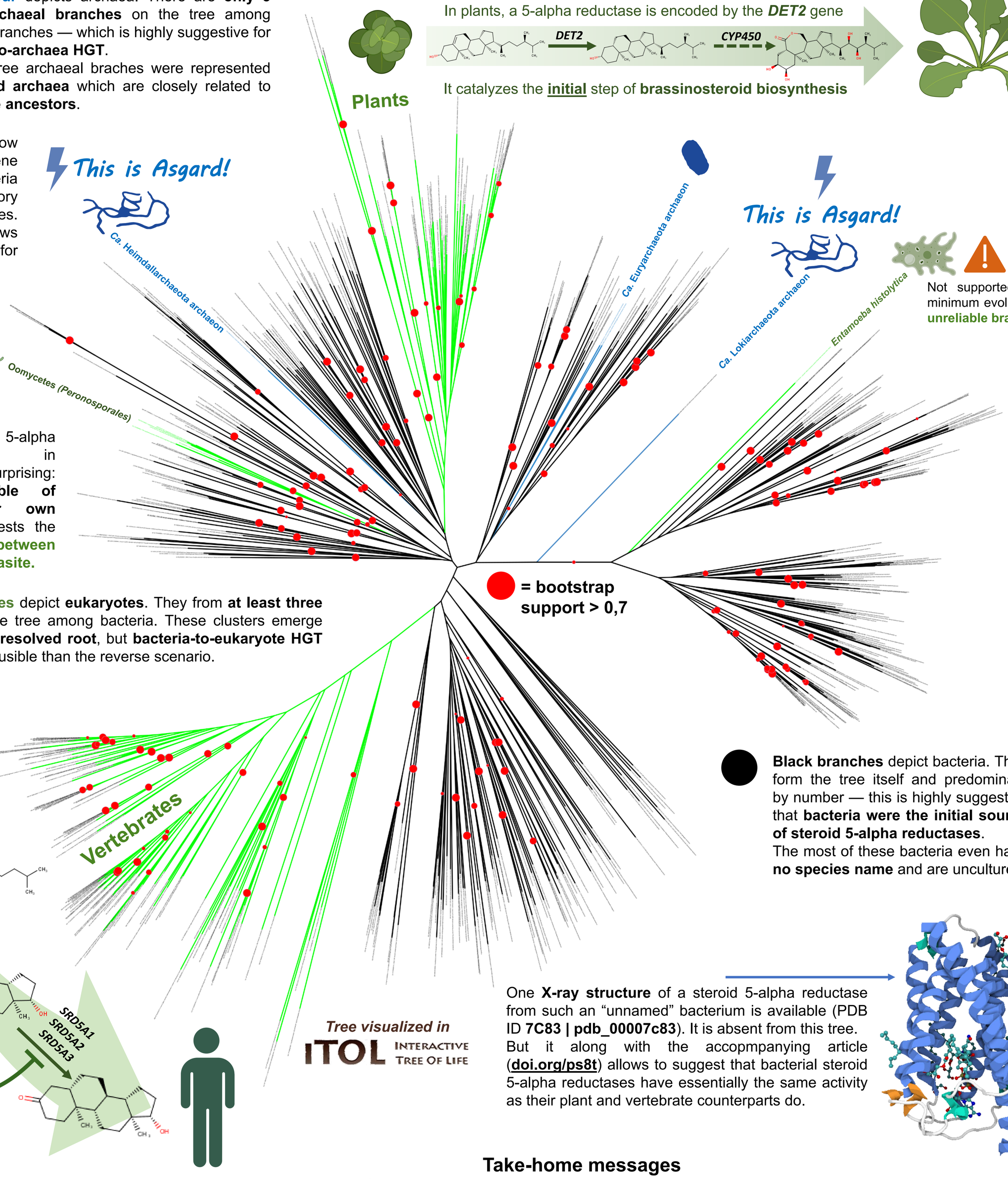
Blue colour depicts archaea. There are **only 3 subtle archaeal branches** on the tree among bacterial branches — which is highly suggestive for **bacteria-to-archaea HGT**.
Two of three archaeal braches were represented by **Asgard archaea** which are closely related to **eukaryote ancestors**.

Archaeal data show that horizontal gene transfers from bacteria occurred in the history of 5-alpha reductases. This indirectly shows their plausibility for eukaryotes.

In plants, a 5-alpha reductase is encoded by the ***DET2*** gene

It catalyzes the **initial** step of **brassinosteroid biosynthesis**





Not supported by the minimum evolution tree, **unreliable branch!**

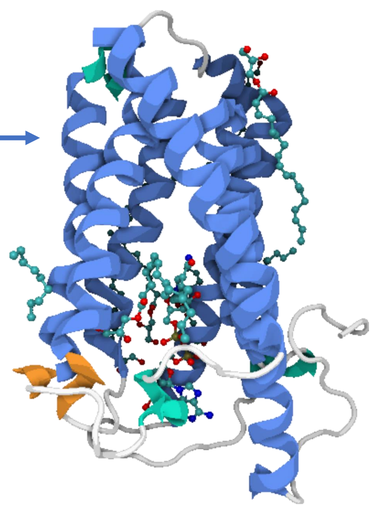
The presence of 5-alpha reductases in ***Peronosporales*** is surprising: they **are incapable of synthesizing their own steroids**. This suggests the **steroid interplay between the host and the parasite.**

Green branches depict **eukaryotes**. They from **at least three clusters** on the tree among bacteria. These clusters emerge from a **poorly resolved root**, but **bacteria-to-eukaryote HGT** is still more plausible than the reverse scenario.

Black branches depict bacteria. They form the tree itself and predominate by number — this is highly suggestive that **bacteria were the initial source of steroid 5-alpha reductases**.
The most of these bacteria even have **no species name** and are uncultured

Tree visualized in iTOL Interactive Tree Of Life

One **X-ray structure** of a steroid 5-alpha reductase from such an "unnamed" bacterium is available (PDB ID **7C83 | pdb_00007c83**). It is absent from this tree. But it along with the accopmpanying article (**doi.org/ps8t**) allows to suggest that bacterial steroid 5-alpha reductases have essentially the same activity as their plant and vertebrate counterparts do.



In vertebrates, 5-alpha reductases catalyze the **final** steps of **male sex steroid** biosynthesis.
In humans, they are encoded by ***SRD5A1***, ***SRD5A2***, and ***SRD5A3*** genes.

Thus, steroid 5-alpha reductases are located completely differently in the steroid hormone biosynthesis of vertebrates and plants.
This increases the plausibility of their independent acquisition by these two groups.

Take-home messages

- The roots of our phylogenetic trees were **poorly resolved**, and the placement of major branches had insufficient bootstrap support. This makes drawing conclusions about the evolutionary history of steroid 5-alpha reductases extremely difficult.
- However, the topology of archaeal and eukaryotic branches makes the scenario of **multiple independent HGTs from bacteria more plausible**. This plausibility increases in the context of the **literature data**.
- I have highlighted some **surprising findings** such as steroid 5-alpha reductases in two **candidate Asgard archaea** and **oomycetes** which do not have their own steroidogenesis. These findings deserve further studies.

**HGT = horizontal gene transfer*

Image credits

Created with BioRender.com

Many thanks to PhyloPic *and especially Mauricio Lucioni (CC BY-NC-SA 3.0)*